

PVLoad-R1 Bench Card

WIRING THE ARDUINO

Ten wires from the Arduino to the 10-pin header on the board. Pin 1 is the square pad.

1	5V	5 V
2	GND	GND
3	D6	K1 drive
4	D7	K2 drive
5	D8	K3 drive
6	D9	CS U2
7	D10	CS U1
8	D11	SDI
9	D12	SDO
10	D13	SCK

On the Uno, 5V and GND are on the power side. D8 to D13 are the six pins nearest the USB socket.

THE THREE BANANA JACKS

J1, red. PV+, the cell's positive.

J2, red. +24 V from the bench supply.

J3, black. Ground. The supply's negative and the cell's negative both go here. Use a stacking plug so they are not jammed in together.

POWER UP

1. 24 V on
2. 5 V on (plug in the Arduino)
3. PV− → J3
4. PV+ → J1
5. illuminate the cell

POWER DOWN

1. darken the cell
2. PV+ out of J1
3. PV− out of J3
4. Arduino out (5 V off)
5. 24 V off

WHAT ON AND OFF MEAN

You never unplug anything from the board to switch a rail. The supply leads stay in J2 and J3. The ten header wires stay on the Arduino.

24 V on and off is the output button on the bench supply. If yours has no output button, put the black lead in J3 first, and pushing the red into J2 is the on step.

5 V on and off is the USB cable, pulled out at the laptop end. Stop the script before you pull it.

USING THE HANDHELD METER

Black lead into the socket marked **COM**. Red lead into the socket marked **VΩ**. Never the **10A** socket: in that one the meter is a plain piece of wire and you would be shorting J1 to J3.

Turn the dial to **ohms** (Ω) to read the board's resistance across J1 and J3. Turn it to **DC volts** (V with a straight line over dashes) to probe pins against ground.

CABLES TO THE LAPTOP

Arduino. Normal USB printer cable, A to B.

EDFA100P laser. Normal USB cable. Nothing special.

Bench supply. Two banana leads to J2 and J3. Set 24.0 V with a current limit of about 100 mA.

The two Keithley 617 meters. These have no USB. They only have a GPIB socket, the wide connector with two screws. You need one USB to GPIB adapter, National Instruments brand. Plug the adapter into the laptop and run a GPIB cable to the first meter. GPIB connectors stack on top of each other, so a second GPIB cable goes from the first meter to the second. Both meters share the one adapter.

Before connecting them, give each meter a different address from its front panel. They both arrive set to 27, which does not work with two on the same cable. Set the volts meter to **22** and the amps meter to **23**.

CONNECTING THE 617 METERS TO THE CELL

The front socket marked INPUT is triaxial, so a banana plug will not fit. Screw on a triax to dual banana adapter. You now have two ordinary banana posts, **HI** and **LO**.

The amps meter goes inside the wire. Current has to flow through it, so cut the wire running from the cell's + terminal to J1 and put the meter in the break.

```
cell +   → HI post
LO post  → J1
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The volts meter goes across the cell. Nothing is cut. Two leads just touch the cell's terminals.

```
HI post  → cell +
LO post  → cell −
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Clip these onto the cell's own leads, not onto J1 and J3. On the jacks you would be measuring through the amps meter as well and the reading comes out low.

USING THE 617 METERS

Power switch is on the front.

On power up the meter sits in Volts with **ZERO CHECK** engaged. While ZERO CHECK is on it reads zero whatever is connected to it. Press the ZERO CHECK button to release it before measuring by hand. The script does this itself when it is driving the meter.

The function buttons pick V, A, ohms or coulombs. The script sets these too.

To set the GPIB address: press SELECT repeatedly until IEEE appears, set the number with the ADJUST arrows, then press SHIFT followed by SELECT to store and exit.

REMOTE lights up while the laptop is driving the meter. Press LOCAL to take manual control back.

THE LASER INTERLOCK

Small connector on the back of the amplifier. It is a safety cutout, meant to be wired to a door switch so the laser dies when the door opens. Plug in the small shorting plug that came with the unit. Left empty, the laser will not fire at all.

LASER SAFETY. Class 3B, 1550 nm. You cannot see it. It puts out up to 30 mW any time it is enabled, at any pump current, even with nothing going into it. Terminate the output. Never look into the fibre.

DO NOT. Never have the cell connected and lit while the 24 V is off. With 5 V up and 24 V missing, the cell's current goes straight into the chip and destroys it. That is why the cell goes on last and comes off first.